```
TYPE=Ethernet
BOOTPROTO=none
HWADDR=52:54:00:01:fa:0a
DEFROUTE=yes
IPV4_FAILURE_FATAL=no
IPV6INIT=yes
IPV6_AUTOCONF=yes
IPV6_DEFROUTE=yes
IPV6_FAILURE_FATAL=no
NAME=ens10
UUID=50c78fec-6902-4b1e-9c79-b7350a204891
DEVICE=ens10
ONBOOT=yes
DNS1=192.168.123.1
IPADDR=192.168.123.13
PREFIX=23
GATEWAY=192.168.123.1
DOMAIN=example.com
IPV6_PEERDNS=yes
IPV6_PEERROUTES=yes
```

Figure 2: Configuration file of interface *ens10*

Correcting network device-naming issues

A network interface device's name can be assigned by setting values in the *ifcfg-** configuration files. The MAC address must be matched with the *HWADDR* variable. The value of the DEVICE variable determines the name of the network interface device corresponding to the specified MAC address. Figure 2 shows the configuration of an interface such as *ens10*.

The screenshot in Figure 3 shows the configuration of an interface from the command line. The following command can be used to view the configuration of the interface from the command line:

```
# ip addr show dev <device-name>
```

Note: Remember that the file name is needed to start with *ifcfg-*, but it does not have to match the specified device's name, although that is a recommended practice. Once the changes are made to the configuration file, the system is rebooted to implement them.

Troubleshooting interface configuration problems

Once network device-naming issues have been resolved, there are other network interface configuration settings that can impair network connectivity. Log files do not provide detailed information about the current network configuration. In that case, the same *ip addr* command mentioned earlier is very helpful, as it displays the active network settings for the network interfaces as shown in Figure 3.

The *ip route* command displays the current routing table. The default route should be checked to ensure it is the correct gateway address for this subnet. An output of the same command has been shown in Figure 4.

```
[root@abc ~]# ip addr show dev ens10
3: ens10: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP qlen 1000
    link/ether 52:54:00:60:81:a1 brd ff:ff:ff:ff:ff:ff
    inet 192.168.123.13/23 brd 192.168.123.255 scope global ens10
       valid_lft forever preferred_lft forever
    inet6 fe80::5054:ff:fe60:81a1/64 scope link
       valid_lft forever preferred_lft forever
[root@abc ~]# █
```

Figure 3: Interface configuration from the command line

```
[root@abc ~]# ip route
default via 192.168.123.1 dev ens10  proto static  metric 100
192.168.122.0/24 dev virbr0  proto kernel  scope link  src 192.168.122.1
192.168.122.0/23 dev ens10  proto kernel  scope link  src 192.168.123.13  metric 100
[root@abc ~]# █
```

Figure 4: Output of the IP route command

If the Dynamic Host Configuration Protocol (DHCP) is used to configure a network interface, then the network manager launches *dhclient* to handle that interface's configuration. Interface network settings, such as the IP address, network mask, the default gateway and DNS servers, are determined by the values provided by the DHCP server. When a network interface uses static addressing, the systems administrator must correctly specify these values.

Inspecting network interface configuration files

Static IPv4 and IPv6 network configuration settings are defined in the */etc/sysconfig/network-scripts/ifcfg-** files. When the network manager is not used, the values in these files determine the persistent settings for network interfaces. In this case, network configuration changes must be made to these files, and then the network should be restarted to apply the changes. A snapshot of the configuration file is given in Figure 2.

The BOOTPROTO variable determines what type of IPv4 configuration is performed on the network interface. A value of 'none' or 'static' indicates that the IPv4 network is statically configured. The *dhcp* or *bootp* values cause *dhclient* to be used to manage the interface configuration. The variables to set for static IPv4 configuration include IPADDR0 and PREFIX0.

For backwards compatibility, the older IPADDR and NETMASK settings are also supported. When the network interface connects to the external network, the GATEWAY variable should have the IP address of the default gateway.

The IPV6INIT variable must have the value 'Yes' for any IPv6 configuration to take place on the network interface. IPV6ADDR specifies the primary static IPv6 address and net mask. Additional IPv6 addresses can be assigned with the IPV6_SECONDARIES variable. The IPV6_AUTOCONF variable determines if IPv6 auto configuration is performed on the network interface.

Managing the network manager configuration settings

When the network manager is used to manage network interfaces on a system, the *nmcli* command can be used to display and change network settings. *nmcli con* show *'CONNECTION_NAME'* is a command that displays the current network configuration for a network manager connection. Values that start with ipv4 or ipv6 correspond to the IPv4 or the IPv6 configuration of an interface, respectively. The Ipv4 and Ipv6 configuration of device *ens10*